Value Screener

What the program does, how it thinks, and the maths behind every number

This guide explains the **financial** side of your screener: where its data comes from, how it scores companies against Warren Buffett's, Charlie Munger's and Peter Lynch's ideas, how news and prices turn into alerts, what "levelling out" means, how the Buy and Sell colours are decided, what to change if you want to take more (or less) risk, and which kinds of companies the three philosophies treat as the safest. The last section writes out every formula.

It describes the program exactly as built on 25 September 2026. All thresholds quoted are the program's defaults; the ones marked *(adjustable)* can be changed on the Settings page.

> **Important.** The screener is a research tool, not investment advice, and it never places trades. Its signals are prompts to look closer. Scores rest on published financial statements and public data, which can be late, restated or wrong. Always read the company's own filings before acting, and consider speaking to a SEBI-registered adviser for personal advice.

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1. What the program is, in one page

Every weekday evening the program does four jobs for the roughly 250 largest listed Indian companies (the Nifty LargeMidcap 250):

1. **Keeps the facts current.** It collects today's official closing prices, the 10-year government bond yield, shareholding and pledge data, lenders' bad-loan ratios and, in results season, fresh financial statements.
2. **Scores each company** three times: how closely it fits Buffett's, Munger's and Lynch's way of picking stocks, each as a percentage. A company "matches" a philosophy only if it scores at least 65% (adjustable) *and* passes that philosophy's non-negotiable "must-pass" tests *and* enough of the data exists to trust the score.
3. **Watches your stocks** (the ones you hold or shortlist) for news-linked price drops that are abnormal for that stock, and tells you later when the price has settled ("levelled out").
4. **Keeps a permanent record** of every recommendation-type moment (shortlisting, alerts, Buy/Sell signal changes, your trades) with the full analysis frozen at that time, so you can later judge how the calls worked out.

> **The core idea.** The three investors agree on a lot: buy good businesses, run by honest people, at sensible prices, and don't overpay. They differ in emphasis. **Buffett** wants durable high returns, little debt and a price below a conservative estimate of value. **Munger** wants an *excellent* business first (very high returns on capital that hold up in bad years) and avoids anything that smells of accounting tricks or weak governance. **Lynch** wants growth he can understand, bought at a price that doesn't already assume the growth (the PEG idea), and he sorts companies into types (fast growers, stalwarts, cyclicals...) because each type is valued differently.

A stock that matches **all three** is a business of high quality, financially safe, reasonably priced and still growing. A stock that matches Buffett and Munger but not Lynch ("quality core") is usually an excellent, mature business. A Lynch-only match is usually a growth story at a fair price but with less proven durability.

2. Where the data comes from

Item	Source	How often
Annual financial statements (10 years)	The Top 250 workbook (built from Screener.in tables), Screener.in public company pages, or your Screener exports	Weekly in results season (May–June, Nov–Dec)
Quarterly results	Screener.in public pages (last 12 quarters)	Same as above
Current share price and market value	NSE's official end-of-day file (the "bhavcopy"); Yahoo Finance if NSE fails or lags	Every weekday
Year-end prices (10 years)	Yahoo Finance	Weekly, only where missing

Promoter %, FII + DII %	Screener.in shareholding table (latest quarter)	Weekly
Promoter pledge %	NSE pledge disclosures	Weekly
Gross NPA % (banks, NBFCs)	Screener.in quarterly results; a few from company results releases	Weekly
10-year G-Sec yield (risk-free rate)	CNBC public quote for India's 10-year bond (7.12% on 25-Sep-2026)	Every weekday
News headlines	Google News	Every weekday, for your stocks
Nifty 50 index (the "market")	Yahoo Finance	Every weekday

No invented numbers. When something isn't published, it stays blank. The score then shows what share of the test could run ("coverage"), and if too little could run, the company can't match. Hover any figure in the app to see its source and date.

3. How a company is scored

Each philosophy is a checklist of tests, each with a **weight** (its importance, in points out of 100).

1. **Partial credit, not pass/fail.** Each test scores between 0 and 1 on a straight line between a "zero marks" point and a "full marks" point. Example: Buffett's debt test gives full marks at debt/equity of 0.5 or less and zero at 1.5; a company at 1.0 gets half marks. This way a near miss is treated as a near miss.
2. **The score** is the weighted average of the tests that have data, as a percentage.
3. **Coverage** is the share of the total weight whose data existed. Below 70% (adjustable) the score isn't trusted and can't match.
4. **Must-pass tests.** Some tests are non-negotiable for a philosophy. If one fails (scores under its minimum), the company does not match however high its total, and if one has no data, the company can't match until the data exists ("needs input").
5. **Red flags (Munger only)** can only subtract points, never add: a clean record is expected, not rewarded.
6. **Banks, NBFCs and insurers** get their own test lists, because debt, capex and free cash flow mean something different for them.

> **Match** = score ≥ 65% (adjustable) **and** coverage ≥ 70% (adjustable) **and** no must-pass test failed **and** no must-pass test missing data.

4. The three philosophies, test by test

"Zero → full" gives the value scoring 0 and the value scoring full marks; values in between score proportionally. "MP" marks a must-pass test with the level it must reach.

4.1 Buffett — "a wonderful business at a sensible price"

Test (ordinary companies)	Weight	Zero → full	Must-pass
ROE ≥ 15% in most of the last 10 years	20	30% of years → 80% of years	MP: at least 55% of years
No loss years	5	any loss → none	
Long-run EPS growth (10y, else 5y)	10	0% → 12% a year	
Low debt / equity	15	1.5 → 0.5	MP: 1.0 or less
Profits turn into free cash (5y FCF ÷ net profit)	15	0.3 → 0.8	
Low capital needs (5y capex ÷ operating cash)	5	1.0 → 0.5	
Interest cover (EBIT ÷ interest)	5	1.5x → 5x	
Margin of safety (DCF value vs market value)	15	−25% → +25%	MP: price at or below value
Each ₹1 retained created ≥ ₹1 of market value (up to 5y)	10	0 → 1.0	

Banks and NBFCs replace debt, cash-flow, capex and interest tests with: return on assets (10; full at 1.2% for banks, 2.5% for NBFCs), Gross NPA (10; zero at 8%, full at 2% or less; **MP: 5% or less**), book value per share growth (15; 3% → 14% a year) and margin of safety on a justified price-to-book (20). **Insurers** use premium growth (20; 0% → 15%), book value growth (15) and the justified P/B margin of safety (20).

4.2 Munger — "an excellent business; avoid stupidity"

Core test (ordinary companies, 100 points)	Weight	Zero → full	Must-pass
Exceptional return on capital (median ROCE, 10y)	25	10% → 25%	MP: median ROCE ≥ 17.5%
Moat held in bad years (worst ROCE, 5y)	10	5% → 15%	MP: worst year ≥ 10%
Stable operating margins (variation of EBIT margin)	10	0.50 → 0.15	
Revenue growth (runway)	10	3% → 12% a year	
Earnings compounding (EPS growth)	10	3% → 15% a year	
Fair price: free-cash-flow yield ÷ G-Sec yield	20	0.3 → 1.0	
Little debt (debt / equity)	15	1.0 → 0.3	

Red flag (subtracts up to)	Points	No deduction → full deduction
Promoter shares pledged	−10	0% → 20% pledged

Persistent dilution (share count growth, 5y)	−8	0.5% → 3% a year
Earnings not backed by cash (operating cash ÷ profit, 5y)	−8	0.9 → 0.6
Receivables growing faster than sales (5y)	−6	0 → 10 percentage points faster
Profit propped up by other income (share of pre-tax profit, 3y)	−6	15% → 35%
Unusually low tax rate (5y)	−4	20% → 10%

Lenders and insurers: high median ROE (25; 10% → 18%; **MP: ≥ 14%**), ROE never collapsed (15; worst 5y ROE 0% → 12%; **MP: ≥ 6%**), book value growth (15), earnings growth (10), Gross NPA for lenders or premium growth for insurers (15), and fair price measured as median ROE ÷ P/B (20; 4% → 10%). Pledge and dilution red flags still apply.

4.3 Lynch — "know what you own, and don't overpay for growth"

Lynch first puts each company into a **category**, then values it accordingly:

- **Cyclical** — earnings rise and fall with the economy or commodity prices: EPS fell more than 20% in two or more of the last five years, or the industry is cyclical (oil & gas, metals, chemicals, cement, autos, realty, textiles) and history is short.
- **Turnaround** — a loss in the three years before the latest, now profitable.
- **Asset play** — cash plus investments exceed half the market value.
- **Fast grower** (EPS growth ≥ 20% a year), **stalwart** (10–20%), **slow grower** (under 10%).

Test	Weight	Zero → full	Must-pass
(Growth % + dividend yield %) ÷ P/E — "PEG" (non-cyclicals)	25	0.5 → 2.0	MP: ratio ≥ 1.0
EPS growth in the 15–25% sweet spot	15	full 15–25%, half 10–15% or 25–35%	
or, for cyclicals: P/E on mid-cycle (median) earnings	25	25 → 10	MP: 20 or less
and can survive a downturn (interest cover)	15	2x → 6x	
EPS rose in most of the last 5 years	10	40% → 80% of years	
Promoter skin in the game	10	25% → 50% held	
Not yet crowded by institutions (FII + DII)	5	60% → 30%	
Size leaves room to grow	5	small 100%, mid 60%, large 30%	
Strong balance sheet (debt / equity)	10	1.0 → 0.35	MP: 0.675 or less
Inventory not piling up (inventory growth − sales growth)	10	+10 points → 0	
Net cash cushion ((cash − debt) ÷ market value)	10	−20% → +10%	

Lenders and insurers replace the last three with book value growth (20) and Gross NPA or premium growth (10).

5. How value and "cheapness" are judged

5.1 Ordinary companies: discounted cash flow (Buffett's margin of safety)

1. Start from **free cash flow** (operating cash flow minus capital spending), averaged over the last 3 years. For cyclicals, the median over all years, so a boom year isn't treated as normal.
2. Grow it for 10 years at the company's own **lower** of its 5- and 10-year EPS growth, capped at 15% and floored at 0%.
3. After year 10, grow it forever at 5% (adjustable), but never above the discount rate minus 2%.
4. Discount everything at **G-Sec yield + 3% equity premium** (adjustable): 7.12% + 3% = 10.12% today.
5. Value of the shares = present value − debt + cash. **Margin of safety** = 1 − market value ÷ this value.

Buffett needs the margin of safety to be zero or better (price at or below value). Full marks come at a 25% discount.

5.2 Banks and NBFCs: justified price-to-book

A lender earning return on equity (ROE) above its cost of equity is worth more than its book value. The program uses the median ROE of the last 5 years, a cost of equity of G-Sec + premium + 1% (extra for leverage: 11.12% today), and growth capped by what the retained profits can fund. Justified P/B = (ROE − growth) ÷ (cost of equity − growth); value = justified P/B × book value.

5.3 Cheapness tests elsewhere

- **Munger:** free-cash-flow yield compared with the G-Sec yield — a business should yield at least what a risk-free bond does.
- **Lynch:** the PEG ratio for growers; P/E on mid-cycle earnings for cyclicals ("a low P/E on peak earnings is a trap").

6. Buy, Sell, In portfolio: the signals and the price trigger

Signal	Rule
Sell	You hold it, and either it matches **none** of the three philosophies, or its price is above your **price-discipline limit** (default: more than 2× the conservative value estimate).
Buy	Shortlisted, not held, matches **at least one** philosophy, and is priced within the limit.
In portfolio	You hold it, it still matches at least one, and it is within the limit.
Watching	Shortlisted, not held, and either not matching or too expensive right now.

A signal can change for two reasons: the **price** moved (the valuation tests re-score every day), or the **business** changed (new results, a pledge, dilution, a rise in bad loans). Every change to Buy or Sell is written to the Recommendation log.

The price trigger

For each held or shortlisted stock the program re-scores the company at many hypothetical prices, keeping everything else fixed, and reports:

- for a stock that is fine now: **the rise at which it would turn to Sell** — the earlier of losing its last match and passing the price-discipline limit;
- for a stock that isn't: **the fall needed** to match again and get back within the limit — its Buy point;
- and, on hover, the same for each philosophy separately.

Price discipline. Munger's only price test ("fair price") carries 20 of about 95 points and is not a must-pass test, so an exceptional business keeps *matching* Munger even when very expensive — faithful to Munger, who held great businesses through expensive periods. Making it a hard must-pass was tested and rejected: it would have removed 39 of 72 Munger matches, because good Indian businesses usually trade at only 1–2% free-cash-flow yields. Instead, the **signals** get a price check while the **scores stay unchanged**: when market value exceeds a multiple of the conservative value estimate (DCF for companies, justified P/B for lenders; 2× by default, adjustable: off, 3×, 2×, 1.5×), a held stock shows Sell and a shortlisted one shows Watching instead of Buy. 2× is the default because the value estimate is deliberately cautious (growth capped at 15%, the lower of the 5- and 10-year rates), so 1.5× would flag most fast growers permanently. On 25-Sep-2026 data, 20 of the 49 Munger-only matches were above 2× (e.g. Asian Paints 2.6×, Britannia 2.4×, Astral 4.7×). If a company has no positive value estimate (e.g. negative free cash flow while it invests heavily), the limit can't be applied and the signal rests on the matches alone.

7. How news is taken into account

1. **Collection.** Each weekday, for every held or shortlisted stock, the program searches Google News for the company's short name (e.g. "Infosys"), its ticker where the ticker is a real word (e.g. ITC), and any extra names you gave it, over the last 4 days.
2. **Relevance filter.** A headline is kept only if its *title* actually names the company. Duplicates are removed.
3. **Tags, not opinions.** Headlines are labelled by keywords: results, merger/acquisition, deal/order, legal/regulatory, governance/management, promoter/pledge, rating/broker. There is **no sentiment scoring**: the program does not decide whether news is good or bad.
4. **Matching news to a trading session.** News published before 15:30 IST belongs to that day's session; after the close, to the next session. All headlines landing on the same session form one "event".
5. **The market decides.** The program then asks whether the stock's price reaction in that session was **abnormally negative** (section 8). If it was, you get an alert listing the headlines. If not, the news is simply recorded.

> News never changes a company's Buffett / Munger / Lynch scores. Scores come only from financial statements, ownership data and price. News drives alerts; the reasoning is that the market's reaction is a better judge of whether news matters than keyword guessing.

A same-session drop shows the market reacting; it doesn't prove that a particular headline caused it. The alert says so.

8. The drop alert: limits and worked example

The method is a standard **market-model event study**, as used in academic finance:

1. **Learn the stock's normal behaviour.** Using up to 120 trading sessions (at least 60) ending 5 sessions before the news, the program fits how the stock usually moves with the Nifty: its **beta** (sensitivity to the market), its **alpha** (drift) and its normal day-to-day noise **σ** (the standard deviation of what the market doesn't explain). The 5-session gap stops pre-news leaks from contaminating "normal".
2. **Measure the surprise.** Over the news session and the next one (2 sessions), it compares what the stock did with what the market model expected. The difference, added up, is the **cumulative abnormal return (CAR)**.
3. **Decide.** A dip is confirmed when the abnormal fall is at least the **larger** of:
   - 3% (adjustable), and
   - 2 × σ × √(sessions) (the "2" is adjustable) — i.e. clearly beyond the stock's own normal noise.
4. Trading volume is reported against normal (flagged if 1.5 times or more), but it doesn't decide the alert.

Worked example. A stock normally has σ = 1.8% of unexplained daily movement and a beta of 1.1. Over 2 sessions the noise hurdle is 2 × 1.8% × √2 ≈ 5.1%, which is larger than the 3% floor, so it needs an abnormal fall of about 5% (≈ 4.96% after converting from log returns). News comes out; over the two sessions the stock falls 7% while the Nifty falls 1%. The model expected about −1.1% (1.1 × −1%), so the abnormal return is about −6%: **a confirmed news-linked dip**, and you're alerted. Had the Nifty fallen 4% the same day, the expected move would be −4.4%, the abnormal fall only about −2.7%, and there would be no alert: the stock merely fell with the market.

A calm stock (σ = 1%) has a noise hurdle of 2.8%, so the 3% floor applies. A volatile one (σ = 3%) needs about 8%. The bar is personal to each stock.

9. Levelling out: how it is decided

After a confirmed dip, the program checks every evening whether the price has **stabilised**. It starts checking once at least 5 sessions have passed after the 2-session drop window. It then looks at the last 5 sessions (adjustable) and needs **all three** of these to be true:

Check	What it means	Default limit
Calm	Recent daily trading ranges (average true range as % of price) are back near what they were before the news	Recent range ≤ 1.25 × the 14 sessions before the news
No drift	The stock is no longer falling (or rising) abnormally relative to the market	|abnormal return over 5 sessions| ≤ 1.0 × σ × √5
Base held	The lowest point since the news is not recent	The low is more than 3 sessions old

When all three hold, you get the "levelled out" alert, with where the price now stands compared with before the news and how deep the low was.

> **What levelling out is not.** It is **not a prediction** that the price will rise, and it doesn't require any recovery: a stock can be "levelled" while still 10% below its pre-news price. It says: the panic (or the news-driven selling) seems to be over, the stock is trading normally again, and this is a sensible moment to *decide* — using the scores and your own judgement — whether the drop was an opportunity or a warning. If it hasn't levelled within 60 trading sessions (about 3 months), the program stops tracking that drop. A newer confirmed dip on the same stock replaces an older one that is still waiting.

10. "How much does it need to go up before I get a Sell?"

There are two separate ideas here, and it helps to keep them apart:

1. **Levelling-out alerts are not sell (or buy) signals.** They don't depend on any percentage rise at all — see section 9.
2. **The Sell signal comes from the scores**, not from news. It appears when a stock you hold stops matching all three philosophies, **or** its price rises above your price-discipline limit (2× the conservative value estimate by default). There is **no fixed percentage**; each stock has its own, and the program now works it out for you as the **price trigger** (section 6), shown on the Scores & shortlist page and on the stock's page. Examples from 25-Sep-2026 data:
   - ITC (0.9× value) matched Buffett and Munger; Buffett's match would be lost at about +13%, Munger's at about +67%, so its Sell point was roughly **+67%**.
   - TCS (0.6× value) matched Buffett and Munger; its quality keeps Munger matching at any price, so its Sell point is the price-discipline limit: about **+215%**, where it would cost 2× its estimated value.
   - Asian Paints (2.6× value) was already above the limit: held, it shows Sell; it would need to fall about 24% to get back under 2×.

A Sell signal does not pop up on your screen. It appears on the dashboard, is written to the Recommendation log, and a held stock turning to Sell is also visible in the daily report's analysis table. (If you'd like a pop-up for Buy/Sell changes, that's a small addition.)

11. If you have a larger (or smaller) risk appetite

All of these are on the Settings page. Save your current settings as a preset first (e.g. "Default"), then experiment; Undo and Reset are always available.

Setting	Default	More risk	Effect
Pass mark	65%	55–60%	More companies match; the extra ones are weaker on some tests.
Equity premium	3%	2–2.5%	Lower discount rate → higher estimated values → bigger margins of safety → more Buffett matches, on thinner cushions.
Long-run growth	5%	6%	Higher values for everything; very sensitive, move in small steps.
Minimum data coverage	0.70	0.60	Scores are trusted with more missing data.
Minimum abnormal drop	3%	2%	Alerts on smaller dips: more "buy-the-dip" candidates, more noise.
Drop size in normal swings (σ)	2.0	1.5	Alerts on drops that are less unusual for the stock.
Calm: recent range ≤ x pre-news	1.25	1.5	"Levelled out" comes sooner, while the stock is still choppier: earlier entry, higher risk of catching a falling knife.
No new low for (sessions)	3	2	Same: an earlier levelling call.
Stop tracking a drop after	60 sessions	90	Waits longer for slow recoveries.
Price discipline	2× value	3× or Off	Lets expensive high-quality stocks stay Buy / In portfolio. Stricter: 1.5×.

Using the screens: for more risk, look beyond "All three" to the **Growth (Lynch)** quick view, the fast-grower and turnaround categories, and mid/small companies. For less risk, stay with **All three** or **Quality core**, raise the pass mark to 70%, raise the equity premium to 4%, and keep the alert defaults.

> Settings change what the screener *flags*, not how risky a company actually is. A lower bar finds more ideas; it doesn't make them better.

12. Which companies are the safest bets under the three philosophies, and why

This is general education about the methodology, not a recommendation of any stock.

Lowest risk: matches all three, or Buffett + Munger ("quality core"), in the stalwart category

- **High, steady returns on capital** (ROCE / ROE well above 15% in most years, and not collapsing in bad years). This is the footprint of a **moat**: something competitors can't easily copy. It means growth can be funded from profits rather than debt or new shares.
- **Little debt** (debt/equity ≤ 0.5, strong interest cover). Debt is what turns a bad year into a crisis. Low-debt companies survive recessions and can buy weaker rivals cheaply.
- **Profits that turn into cash** (operating cash ≥ 90% of profit, healthy free cash flow). This is the best defence against accounting tricks.
- **Clean governance**: no promoter pledge, no persistent dilution, no reliance on other income or an unusually low tax rate.
- **Stalwarts** (10–20% earnings growth) in **large** companies: slower than fast growers but far more predictable.
- **Bought with a margin of safety**: even great businesses become poor investments at too high a price; the cushion protects you if your estimate of value is too optimistic.

Among lenders, the equivalent is: Gross NPA under 2%, steady ROE of 15% or more, book value compounding, and a price not far above justified book.

Higher risk (not necessarily bad, but needs more care)

- **Cyclicals** (metals, oil & gas, cement, autos, chemicals): earnings swing; they look cheapest at the top of the cycle.
- **Turnarounds**: the recovery may not last.
- **Fast growers at high P/E** (PEG below 1): the price already assumes years of growth; any stumble is punished.
- **High debt**, **pledged promoter shares**, or **short histories** ("provisional" in the app).
- **Asset plays**: the value may never be unlocked.

Two further safety habits the philosophies share: **spread across sectors** (use the Sector filter so one industry's bad year doesn't dominate), and **know why you own it** (Lynch's rule: be able to explain the story in two minutes).

13. Known limits

- Financial statements are summaries; one-off items, accounting changes or restatements can distort a year.
- The workbook and Screener figures are rounded; tax is recomputed from a rounded tax rate (differences up to ~2%).
- The DCF is deliberately simple and conservative; it will undervalue some high-growth or asset-light businesses and overvalue some cyclicals.
- Ownership, pledge and NPA figures are quarterly and can be up to three months old.
- News coverage from Google News is broad but not complete, and relevance is judged from headlines only.
- The drop test uses the Nifty 50 as "the market"; a whole-sector sell-off can show up as abnormal for individual stocks.
- Past returns on capital don't guarantee future ones; moats erode.

14. Appendix: the maths

14.1 Building blocks

Equity = share capital + reserves · Capital employed (CE) = equity + borrowings · EBIT = profit before tax + interest

Averages use the start and end of the year: $\text{avg}(X) = (X_t + X_{t-1}) \div 2$

ROE = net profit ÷ avg equity · ROA = net profit ÷ avg total assets · ROCE = EBIT ÷ avg CE

ROIC = $\text{EBIT} \times (1 - t) \div \text{avg}(\text{CE} - \text{cash})$, t = that year's tax ÷ profit before tax, kept within 0–40% (25.17% if pre-tax profit ≤ 0)

EPS = net profit (attributable to owners) ÷ shares · Book value per share = equity ÷ shares

Capex = reported purchase of fixed assets; if not reported, Δ net block + Δ capital work in progress + depreciation

Free cash flow (FCF) = cash from operations − capex

Debt/equity = borrowings ÷ equity · Interest cover = EBIT ÷ interest

Market value (market cap) = latest closing price × shares outstanding

P/E = market cap ÷ net profit (sum of the last 4 quarters if they are newer than the last annual report) · P/B = market cap ÷ equity

Dividend yield = dividends ÷ market cap · Payout = dividends ÷ net profit

14.2 Growth rates

$\text{CAGR} = (\text{last} \div \text{first})^{1/\text{years}} - 1$

"5-year" growth uses up to the last 5 years but needs at least 3 years between the endpoints; "10-year" needs at least 6. If the latest value is a loss, growth is set to −100% (fails); if the starting value was a loss, growth is set to 0% (no credit for recovering from a loss).

14.3 Quality ratios

Cash conversion = Σ operating cash (5y) ÷ Σ net profit (5y) · FCF conversion = Σ FCF ÷ Σ net profit (last 5y, ≥ 3 years)

Capex intensity = Σ capex (5y) ÷ Σ operating cash (5y) · Other-income share = Σ other income (3y) ÷ Σ profit before tax (3y)

Average tax rate = Σ tax (5y) ÷ Σ profit before tax (5y) · Margin stability = standard deviation ÷ mean of EBIT margin (up to 10y, ≥ 4 years)

Retained-earnings test = $(P_1 \times S_1 - P_0 \times S_0) \div \Sigma$ (net profit − dividends), over up to 5 years (P = year-end price, S = shares)

14.4 Scoring

linear(x; zero, full) = clamp((x − zero) ÷ (full − zero), 0, 1) — works whether "full" is above or below "zero"

Score = $100 \times \Sigma$ ($weight_i \times s_i$) ÷ Σ $weight_i$ over tests with data

Coverage = Σ weight (tests with data) ÷ Σ weight (all core tests)

Munger: Score = $\max(0, \text{core score} - \Sigma\ (1 - s_{flag}) \times \text{max-penalty}_{flag})$

Match ⇔ Score ≥ pass mark ∧ Coverage ≥ min coverage ∧ every must-pass s ≥ its minimum ∧ every must-pass has data

Must-pass minimums in plain numbers: Buffett — ROE ≥ 15% in ≥ 55% of years; debt/equity ≤ 1.0; margin of safety ≥ 0; Gross NPA ≤ 5% (lenders). Munger — median ROCE ≥ 17.5% and worst 5y ROCE ≥ 10% (lenders: median ROE ≥ 14%, worst ≥ 6%); Gross NPA ≤ 5%. Lynch — PEG ratio ≥ 1.0 (cyclicals: mid-cycle P/E ≤ 20); debt/equity ≤ 0.675; Gross NPA ≤ 5%.

14.5 Valuation

Discount rate r = G-Sec yield + equity premium · g_1 = clamp(min(EPS CAGR 5y, EPS CAGR 10y), 0, 15%) · g_T = min(long-run growth, r − 2%)

$CF_t = FCF_{base} \times (1 + g_1)^t$, t = 1…10 · PV = Σ $CF_t \div (1 + r)^t + [CF_{10} \times (1 + g_T) \div (r - g_T)] \div (1 + r)^{10}$

Equity value = PV − debt + cash · Margin of safety = 1 − market cap ÷ equity value

Lenders: $k_e = r + 1\%$ · g = min(ROE × (1 − payout), book value CAGR, $k_e - 2\%$) · justified P/B = (ROE − g) ÷ (k_e − g) · value = justified P/B × equity

Munger fair price = ($FCF_{3y\ avg}$ ÷ market cap) ÷ G-Sec yield · Lenders: median ROE ÷ P/B

Lynch PEG ratio = (EPS growth % + dividend yield %) ÷ P/E · Cyclicals: normalised P/E = price ÷ median EPS (up to 10 years)

14.6 Price trigger

For factor f, rescale price and market cap by f and re-score. Sell point = smallest $f > 1$ at which no philosophy matches (searched up to 10×); Buy point = largest $f < 1$ at which one starts to match (searched down to 0.2×). Found by bisection to about 0.01%.

Price discipline: multiple = market cap ÷ value estimate; overvalued ⇔ limit > 0 and multiple > limit. Its trigger factor is f = limit ÷ multiple (the value estimate doesn't depend on price). Sell point = min(match-loss factor, f); Buy point = min(match-start factor, f).

14.7 Drop alert (event study)

Log return: $r_t = \ln(P_t \div P_{t-1})$ using dividend-adjusted closes; m_t = same for the Nifty 50

Fit $r_t = \alpha + \beta\, m_t + \varepsilon_t$ by least squares over up to 120 sessions ending 5 sessions before the event (≥ 60 needed)

$\beta = \Sigma\,(m - \bar{m})(r - \bar{r}) \div \Si\,(m - \bar{m})^2$ · $\alpha = \bar{r} - \beta\,\bar{m}$ · $\sigma = \sqrt{(\,\Sigma\,\varepsilon^2 \div (n - 2)\,)}$

Abnormal return $AR_t = r_t - (\alpha + \beta\, m_t)$ · $CAR = \Sigma\, AR_t$ over the event session and the next (n = 2)

Dip confirmed ⇔ $CAR \le -\max(\,-\ln(1 - 3\%),\, 2 \times \sigma \times \sqrt{n}\,)$ · z-score = CAR ÷ $(\sigma\sqrt{n})$ · reported $\% = e^{CAR} - 1$

Volume ratio = mean volume in the window ÷ median volume in the estimation period (flag ≥ 1.5)

14.8 Levelling out

True range % = max(high − low, |high − prev close|, |low − prev close|) ÷ prev close (dividend-adjusted)

Calm ⇔ mean TR% (last 5 sessions) ≤ 1.25 × mean TR% (14 sessions before the event)

No drift ⇔ $|\,\Sigma$ AR (last 5 sessions)$\,| \le 1.0 \times \sigma \times \sqrt{5}$

Base held ⇔ the lowest dividend-adjusted low since the event is more than 3 sessions old

Levelled ⇔ calm ∧ no drift ∧ base held, checked once ≥ 5 sessions have passed after the 2-session window; tracking stops after 60 sessions